

PROJECT HANDOUT

AttendEase

A role-based college attendance management system built with the MERN stack for The Kavery Engineering College (Autonomous), Mecheri. This handout summarizes the project scope, system modules, technical architecture, and the professional build prompt.

CATEGORY	STACK	RELEASE CONTEXT
College ERP / Attendance Portal	MongoDB, Express, React, Node.js	CSE Department Focused Build

Project Overview

AttendEase is a secure academic attendance platform that centralizes student attendance, subject management, schedules, and reporting across multiple user roles. It is designed to give administrators full operational control, allow teachers to mark and update attendance efficiently, provide students with transparent attendance visibility, and help HOD or Principal users monitor compliance through summary reports.

Role Modules

Admin - Manage teachers, subjects, students, and schedules - Maintain CSE-only department configuration - Handle student create, edit, and delete operations - Assign faculty to subjects and academic timetables	Teacher - Load students by department, year, semester, and section - Mark attendance as Present, Absent, or Late - Use bulk actions and keyboard shortcuts for speed - Update attendance records for the same class date
Student - View profile and personal attendance percentage - Track subject-wise attendance performance - Identify low attendance below the 75% threshold - Refresh live report data from the system	HOD / Principal - Monitor overall monthly attendance reports - Review subject-wise and semester-wise summaries - Detect low-compliance classes and trends - Use management-facing attendance tables for review

System Architecture

Frontend - React 18 with Vite and React Router - Welcome page, role selection, login flow, protected dashboards - Reusable dashboard layout and API service layer - Session persistence using localStorage	Backend - Express API with MongoDB and Mongoose - JWT authentication and bcrypt password hashing - Role-based middleware and centralized error handling
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- Attendance, reports, schedules, notifications, and requests APIs

Key Functional Expectations

- JWT-based login with role validation and protected route access.
- Attendance percentage calculation where Present and Late count as attended.
- Duplicate prevention for the same student, subject, and date.
- Monthly reporting and low-attendance monitoring below 75%.
- PDF and Excel export support for management reports.
- Professional academic UI suitable for institutional daily use.

Professional Development Prompt

Build a full-stack web application called "AttendEase", a professional college attendance management system for The Kavery Engineering College (Autonomous), Mecheri, using the MERN stack: MongoDB, Express.js, React.js with Vite, and Node.js.

Create a secure, role-based platform that supports Admin, Teacher, Student, and HOD / Principal users. The system must include a college-branded welcome page, role-selection login flow, protected dashboards, and session persistence using localStorage.

Admin features must include CSE-only department management, teacher management, subject assignment, student CRUD operations, and academic schedule management. Teacher features must include filtered student loading, attendance marking, status updates, bulk marking, and keyboard shortcuts. Student features must include profile view, overall attendance percentage, subject-wise summary, and low-attendance highlighting below 75%. HOD / Principal features must include monthly overall attendance reporting, subject-wise summaries, semester-wise summaries, and management-ready report tables.

Implement the backend with Express.js and MongoDB using models for User, Student, Department, Subject, Attendance, Schedule, Notification, and AttendanceRequest. Add JWT authentication, bcrypt password hashing, role-based authorization, centralized error handling, and REST APIs for login, profile, catalog data, student management, attendance, reports, schedules, notifications, and institutional management.

Use a clean, responsive academic interface with structured navigation, reliable validation, reusable frontend service helpers, and a maintainable project structure. The final system should feel secure, professional, practical, and deployment-ready for college attendance operations.

Deliverables

- Complete frontend and backend implementation
- MongoDB schema and API structure
- Role-based login and dashboard workflow
- Attendance marking, tracking, and reporting modules
- Printable documentation and project setup instructions

Prepared from the current AttendEase codebase and project documentation on 25 April 2026.